AP CALCULUS AB

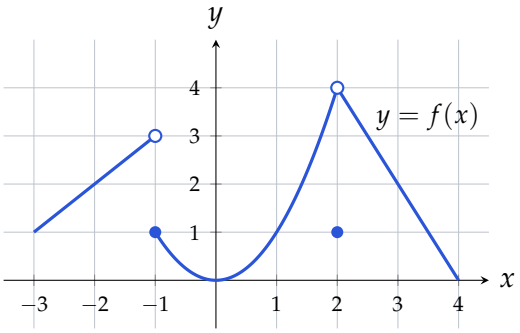
Unit 1 • Lesson 1.3 practice • Answer key

Lesson 1.3 • Estimating Limit Values from Graphs

5 multiple-choice questions and 1 free-response question • Answer key with explanations

Directions

Questions 1 to 5 are multiple choice: choose the best of the four answer choices. Question 6 is free response: show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.



The graph of f: a line segment for $-3 \le x < -1$, $y = x^2$ for $-1 \le x < 2$, a line segment for $2 < x \le 4$; $f(-1) = 1$, $f(2) = 1$.

Question 1

NO CALCULATOR

Using the graph of f, what is $\lim_{x \to -1^-} f(x)$?

(A) 3

(B) 1

(C) 2

(D) It does not exist.

Solution

Answer: (A). From the left, the line segment approaches the open circle at height 3.

Question 2

NO CALCULATOR

Using the graph of f, what is $\lim_{x \to 2} f(x)$?

(A) 1

(B) 2

(C) 4

(D) It does not exist.

Solution

Answer: (C). Both pieces approach the open circle at $(2, 4)$.

Question 3

NO CALCULATOR

Using the graph of f, what is $f(2)$?

(A) 4
(B) 1
(C) 0
(D) undefined

Solution
Answer: (B). The filled dot at $x = 2$ is at height 1.

Question 4

NO CALCULATOR

For which value of x in $(-3, 4)$ does $\lim f(x)$ fail to exist?

(A) $x = 2$
(B) $x = 0$
(C) $x = 3$
(D) $x = -1$

Solution
Answer: (D). At $x = -1$ the one-sided limits are 3 and 1.

Question 5 (graph)

NO CALCULATOR

Use the graph of f above. What is $\lim_{x \to -3^+} f(x)$?

(A) 1
(B) -3
(C) 0
(D) It does not exist.

Solution
Answer: (A). The graph starts at the filled point $(-3, 1)$; as x approaches -3 from the right, $f(x) = x + 4$ approaches 1.

Question 6 (Free response)

NO CALCULATOR

Use the graph of f above.

(a) Find $\lim_{x \to 3} f(x)$ and $\lim_{x \to 0} f(x)$.
(b) Find $\lim_{x \to 1} \big(f(x) + 2x\big)$.
(c) Find $\lim_{x \to 2} x\, f(x)$.

Solution
(a) $\lim_{x \to 3} f(x) = 8 - 6 = 2$ and $\lim_{x \to 0} f(x) = 0^2 = 0$.
(b) $\lim_{x \to 1} f(x) = 1$ and $\lim_{x \to 1} 2x = 2$, so the limit is 3.
(c) $\lim_{x \to 2} x = 2$ and $\lim_{x \to 2} f(x) = 4$, so the limit is 8.